Diego Atencia

AI Engineer — Applied GenAI for Banking · LLMs · RAG · AWS

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Summary

AI engineer who takes GenAI from idea to production: built and shipped regllm.xyz end-to-end (LoRA-fine-tuned Qwen2.5-7B, RAG pipeline, full AWS deployment) and automated credit-risk model reporting with Copilot Studio inside a regulated bank. Deep banking domain — IRB/LGD modeling and regulatory data (FINREP, COREP, CIRBE) — on top of an M.Sc. in AI and a B.Sc. in Mathematics (UAM). Open-source contributor to LLM training infrastructure (Liger-Kernel, TorchRL).

Production GenAI

regllm.xyz — GenAI app for Spanish banking regulation Q&A *Production, 2026*

- Sole builder, idea to production: fine-tuned Qwen2.5-7B with LoRA on regulatory corpora; RAG pipeline with PostgreSQL/pgvector; full-stack AWS deployment (ECS Fargate, RDS, ALB).
- Live in early access, serving real user queries (~30 to date) — not a PoC.

Experience

Data Scientist — Credit Risk (IRB Models) *Nov 2023 – Present*

[Company] — Madrid

- Built and validated LGD (Loss Given Default) models for IRB credit-risk portfolios, ensuring regulatory compliance.
- Put AI automation into production inside a bank: Power Automate/Copilot Studio workflows integrating LGD model outputs into structured Word reporting pipelines used by the risk team.

SQL & PL/SQL Developer *Oct 2022 – Nov 2023*

Álamo Consulting — Madrid

- Owned regulatory reporting pipelines (FINREP, COREP, SIRBE/CIRBE) for an external banking entity using Oracle and T-SQL.
- Built scalable transformations within a large regulatory codebase, ensuring accurate and timely data delivery.

Open-Source Contributions (LLM & RL Infrastructure)

Liger-Kernel — github.com/linkedin/Liger-Kernel PR #1128 (under review)

Production CUDA kernel library for LLM training, used at scale across industry

- Debugging a distributed-training bug in `LigerTiledSwiGLUMLP`: FSDP post-backward reshard hooks fired prematurely inside a tiling loop, corrupting parameter shards across GPUs; adapting axolotl's gradient-accumulator approach as the fix.

TorchRL — github.com/pytorch/rl Discussion #3538

- Proposing and implementing MC-PILCO: model-based RL with Gaussian Process world models and Monte Carlo rollouts — directly relevant to decision systems and optimization under uncertainty.

LEANN Issue #41

- Proposed a hybrid BM25+RRF retrieval pipeline over a neural-only architecture to improve recall on sparse queries.

Other Projects

Hand Prosthesis Control with Deep RL — blog *M.Sc. thesis*

- Multimodal RL agent (PPO/DDPG) integrating stereoscopic vision and sensor data for real-time prosthetic hand control in MuJoCo simulation.

Education

M.Sc. in Artificial Intelligence (8.67/10) *Sep 2024 – Oct 2025*

International University of Valencia

Thesis: *Optimizing low-cost hand prosthesis control using Deep Reinforcement Learning.*

B.Sc. in Mathematics *Sep 2018 – Jun 2023*

Autonomous University of Madrid (UAM)

Final thesis: Weak solutions to Navier–Stokes equations (9/10). Founder of university AI student association (70+ members).

Technical Skills

ML / GenAI	PyTorch, HuggingFace, LoRA/QLoRA, RAG pipelines, TensorFlow, Scikit-learn; RL (PPO/DDPG, model-based)
Languages	Python (expert), SQL (expert), C/C++ (strong)
Cloud & Systems	AWS (ECS Fargate, RDS/pgvector, ALB), Docker, Linux, Git/GitHub, CI/CD; distributed training (FSDP/DDP), CUDA kernels
Enterprise AI	Power Automate, Copilot Studio, document/reporting automation in regulated environments
Databases	Oracle, PostgreSQL, ChromaDB (vector), SAS

Languages

Spanish (native)   English (fluent, professional working proficiency)   Mandarin Chinese (learning)